With recombinant DNA technology it is possible to mutate a gene at specific positions in the test tube by SITE SPECIFIC MUTAGENESIS and then introduce the mutated gene back into the cell and investigate the phenotypic changes produced by the mutation in vivo. Such techniques enable geneticists to study, for example, genes with unknown function and specific sequences involved in regulating a gene's expression.

12.7 Transfer of Genetic Material

Normally, genes and the characteristics they code for are passed down from parent to progeny. This is called vertical gene transfer. Bacteria and some lower eukaryotes are unique in that they can pass DNA from one cell of the same generation to another. The exchange of genes between two cells of the same generation is referred to as horizontal gene transfer. Mechanisms like transformation, transduction and conjugation takes place naturally and may bring about genetic variation and genetic recombination. These gene transfer mechanisms are also employed in genetic engineering to introduce desired gene into the cells. Introducing a foreign gene or recombinant DNA into the cells is one of the techniques used in genetic engineering. The success of cloning depends on the efficiency of gene transfer process. The most commonly employed gene transfer methods are transformation, conjugation, transduction, electroporation, lipofection and direct transfer of DNA. The choice of the method depends on the type of host cell (bacteria, fungi, plant, animal). Figure 12.15 shows methods of DNA transfer.

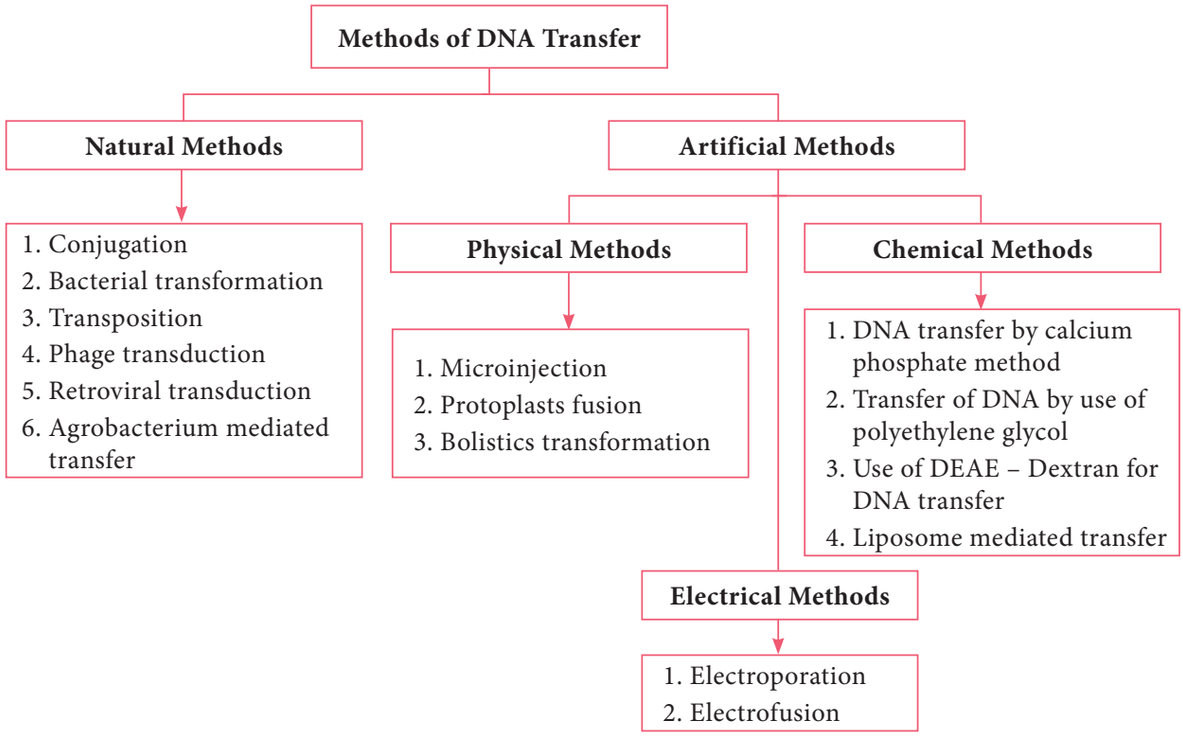


Figure 12.15: Methods of DNA transfer

Note: The term Transfection is used for the transfer of DNA into eukaryotic cells by various physical or chemical means.

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